

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 14

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Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!
- Quiz on Friday on polar coordinates and beginning of triple integrals
- Exam I – One-Page Correction (+5) – due 3/9 – can be found on HuskyCT > Quizzes and Exams (will not be accepted late)

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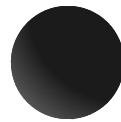


ALVARO: Start the recording!



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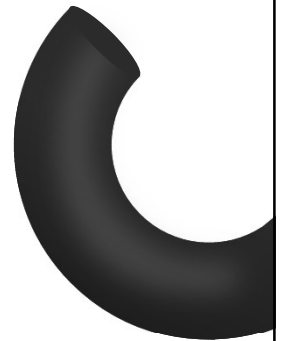
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More on Triple Integrals in
Cylindrical Coordinates



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Today – Triple Integrals in Cylindrical Coordinates

- Cylindrical Coordinates
- Triple Integrals in Cylindrical Coordinates

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Cylindrical Coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z \quad r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad z = z$$



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Example: Evaluate the value of the triple integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$



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[Extra]

Example: Evaluate the value of the triple integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$

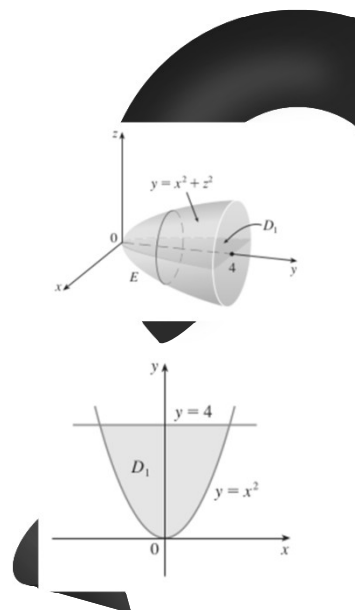


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Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



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Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

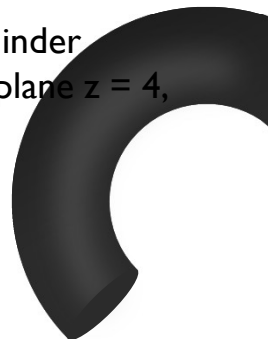
where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.

[Extra]



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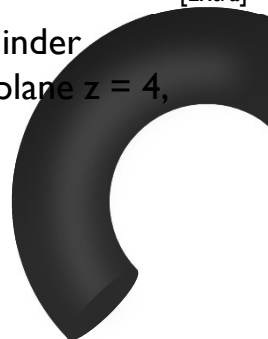
Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



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[Extra]

Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



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Questions?

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ALVARO: Start the recording!



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Triple Integrals in Spherical Coordinates

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Today – Triple Integrals in Spherical Coordinates

- Spherical Coordinates
- Triple Integrals in Spherical Coordinates

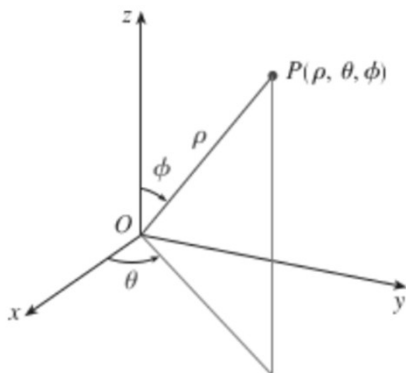
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Spherical Coordinates



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Spherical Coordinates



$$x = \rho \sin \phi \cos \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \phi$$

$$\rho^2 = x^2 + y^2 + z^2$$



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Cylindrical vs Spherical Coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z \quad x = \rho \sin \phi \cos \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \phi$$

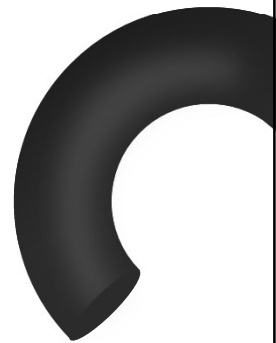
$$r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad z = z$$

$$\rho^2 = x^2 + y^2 + z^2$$



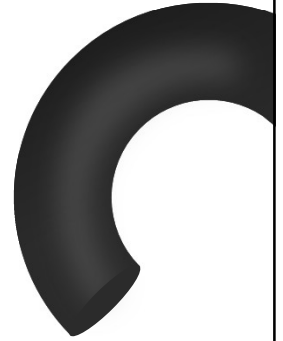
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Example: Sketch the surface given by $\rho = k$.



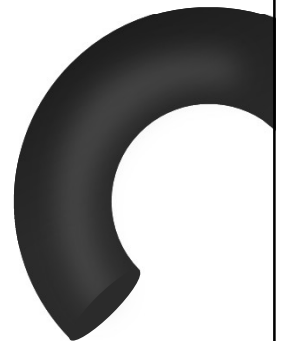
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Example: Sketch the surface given by $\theta = k$.



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Example: Sketch the surface given by $\phi = k$.



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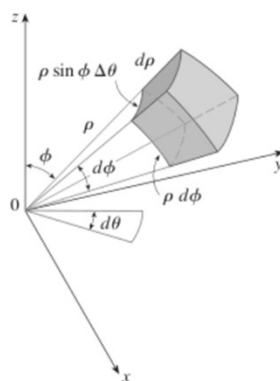
Example: Sketch the “spherical wedge” given by
 $a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d.$



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Triple Integrals in Spherical Coordinates

Volume element in spherical coordinates: $dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$



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Triple Integrals in Spherical Coordinates

$$\begin{aligned} \iiint_E f(x, y, z) \, dV \\ = \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \end{aligned}$$

where E is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$$

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Example: Use triple integrals to find the volume of the sphere of radius R .

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[Extra]

Example: Use triple integrals to find the volume of the sphere of radius R.

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Example: Evaluate

$$\iiint_E e^{(x^2+y^2+z^2)^{3/2}} dV$$

where E is the unit sphere/ball.

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[Extra]

Example: Evaluate

$$\iiint_E e^{(x^2+y^2+z^2)^{3/2}} dV$$

where E is the unit sphere/ball.

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Example: Use triple integrals to find the volume of the “ice-cream cone” that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $z = x^2 + y^2 + z^2$.

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[Extra]

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[Extra]

Example: Use triple integrals to find the volume of the “ice-cream cone” that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $z = x^2 + y^2 + z^2$.



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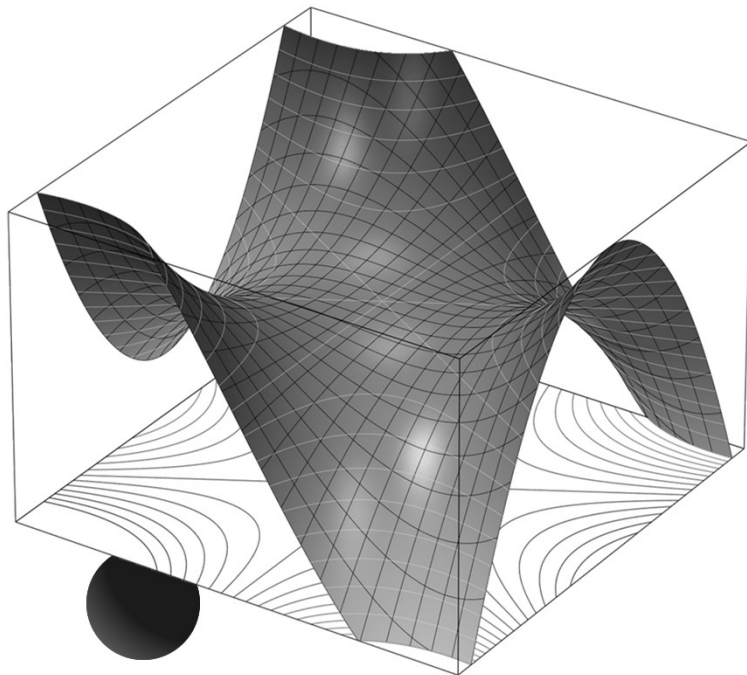
Questions?



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Thank you

Until next time.



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